



Exercise 05: PID Control

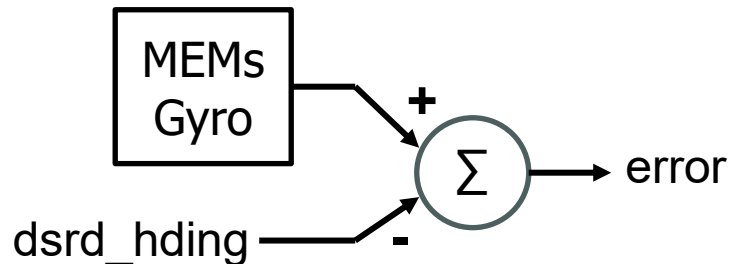
- **PID** is a very common control scheme.
 - More info: <https://www.youtube.com/watch?v=fv6dLTEvI74>

- Project this semester: MazeRunner
 - Needs to control where it goes (90° turns, equi-distance from the walls)



PID in MazeRunner:

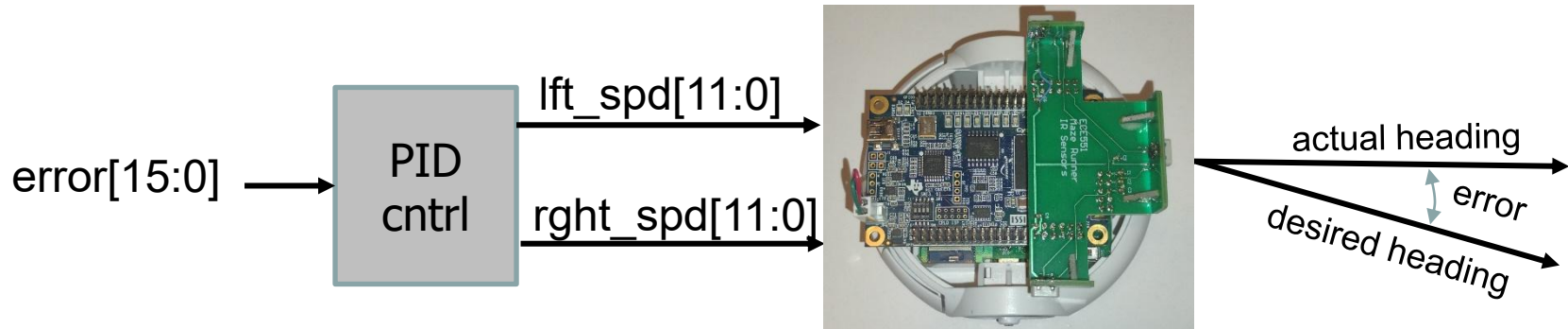
- Want the MazeRunner to follow a heading



- Control knobs: duty cycle to the left & right motors used to steer the MazeRunner.
- Need to convert this heading error into left/right motor speed control



Exercise 05: PID Control



- Job of PID unit is to drive error to zero
 - Convert error signal to motor speed signals
- In this example, measured heading is counter clockwise from desired heading (i.e., error is +ve) → want to drive left motor harder than right
- Each motor will have a forward component to its duty cycle + a component added in by the PID block to drive the error to zero.
 - Component will be added to left motor, subtracted from right motor
- Determining how far we “turn this control knob” is the job of the PID block.



PID – P for Proportional

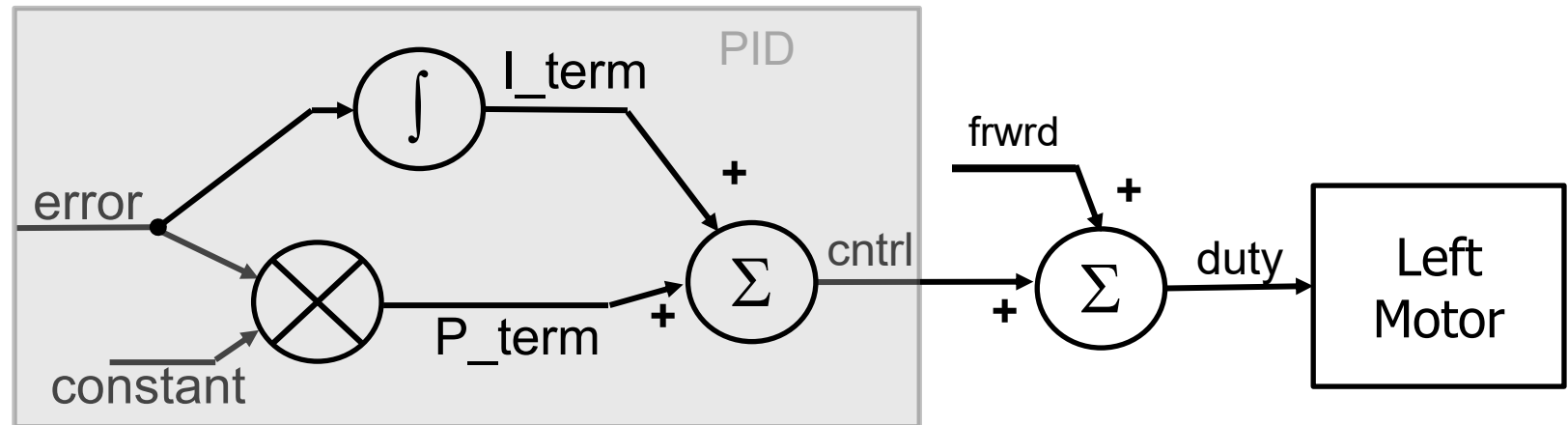
- Consider just the left motor
 - One component will be proportional to the error
→ if the error is +ve, we need to drive the left motor harder to correct
This component would be the **P** term (**P**roportional) of the **PID**.
- Can we just use the P term alone in our system?

Consider these questions:

- Are the two motors of the MazeRunner exactly matched?
 - If you gave them the same duty cycle would they run at the exact same speed?
- If we only have the P term, $\text{control} = \text{constant} * \text{error}$
If $\text{error} = 0$, then $\text{control} = 0$... how do we compensate for motors that are not perfectly matched?
- Proportional control alone cannot drive **error** to zero
 - We need to **integrate** error over time and use this to drive the error to zero



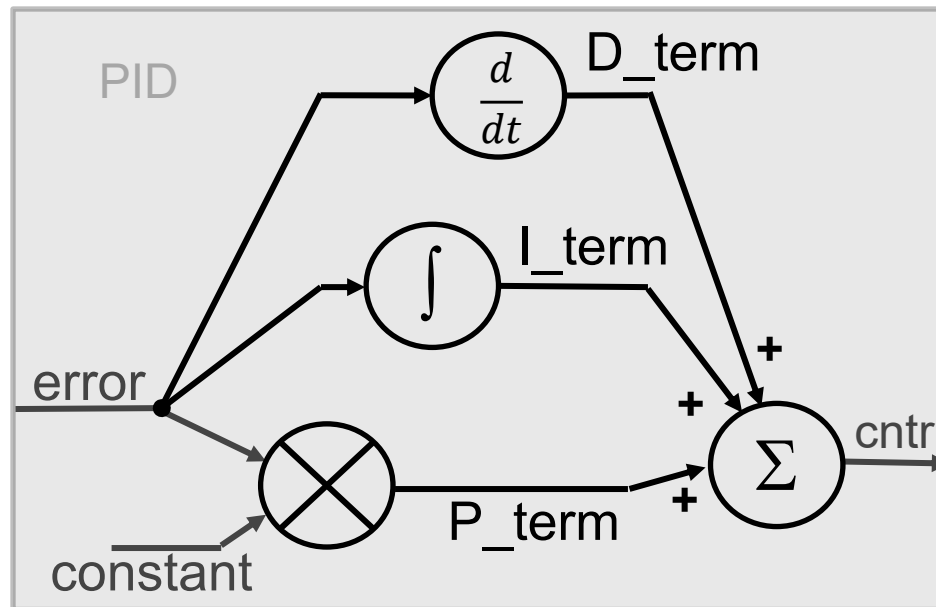
PID – I for Integral



- If **error** is +ve and non-zero then the **I_term** ramps up.
- As the error approaches zero the **P_term** is backing off and approaches zero.
- If left motor is weaker, **I_term** would ramp up to provide a non-zero boost to the duty cycle of the left motor to compensate.
→ This allows the error term to be driven to zero
- One common problem: many systems have lag...
 - The response of the system is not immediate to a change in the control variable.
- By the time **error** hits zero the **I_term** might have ramped up too far (overshoot)...



PID – D for Derivative



- Use the time rate of change (*derivative*) of the **error** term to keep your control scheme from developing too much "momentum" and overshooting your desired target



Exercise 05: PID Control

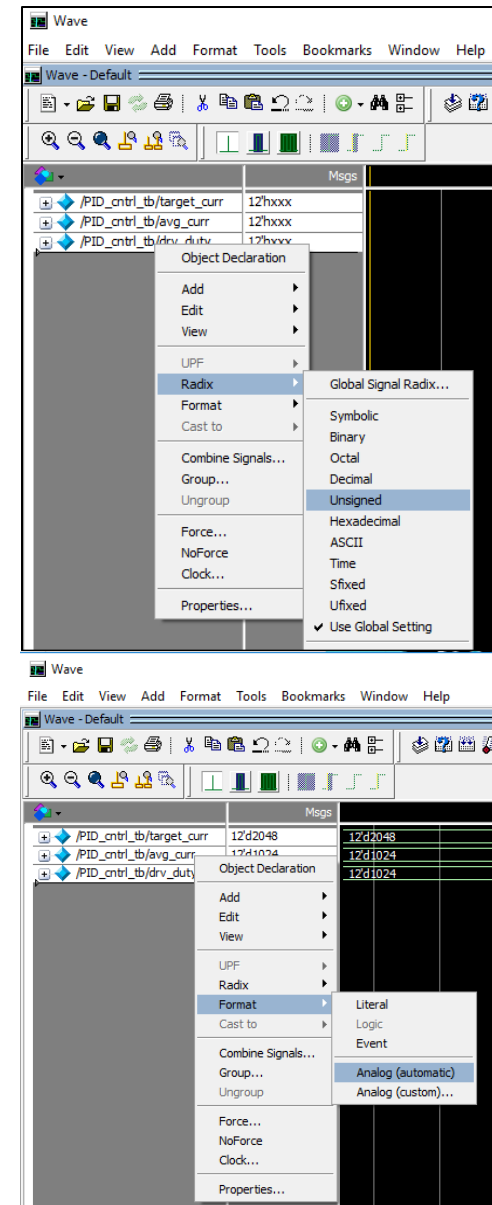
- 3 Verilog files provided for this exercise:
 - **plant.sv** → Controls folks often refer to the system being controlled as the plant.
 - **PID_cntrl.sv** → You will modify a few constants in this PID controller to see the effects
 - **PID_cntrl_tb.sv** → testbench that you will run
- Download these files and make a ModelSim project from them.
- Take a few minutes to study the implementation of **PID_cntrl.sv**
- Simulate the testbench (**PID_cntrl_tb.sv**)
- View the signals: **error**, **heading_deviation**, and **drv_duty** as analog waves



Exercise 05: PID Control

To view waveforms as analog:

- Select the signals of interest and change the radix. Choose **Unsigned** for unsigned signals, and **Decimal** for signed signals.
 - Choose *Decimal* for **error**, and *Unsigned* for **drv_duty** and **heading_deviation**.
- If you have not already... run the simulation
- Right click on the signals again and change the display format to "**Analog (automatic)**"
 - It can't automatically scale signals for which it does not have data, so you have to have run the simulation to do this.





Exercise 05: PID Control

- **PID_cntrl.sv** as given has zero coefficient for both **I** and **D** terms. What is the steady state error with just a P term?
- What happens if you increase the value of P_coeff?
- Now add in a little bit of I term (pick a small value for I_coeff). What do you see?
 - Submit a screen capture of this waveform
- Can you improve this performance? If you see some overshoot and slight oscillations, they can be improved by adding in some D term.
 - Submit an image of your improved response with D term.